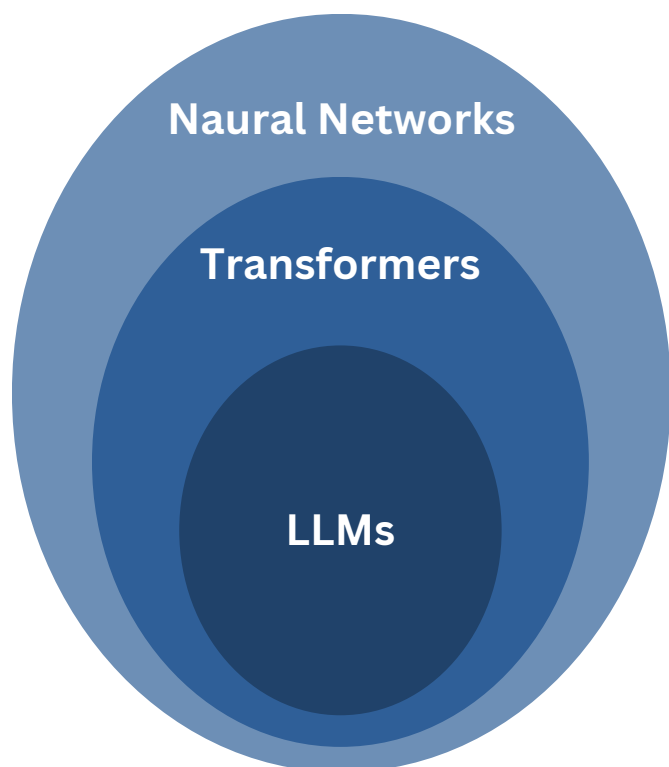


Introduction to AI Product Management

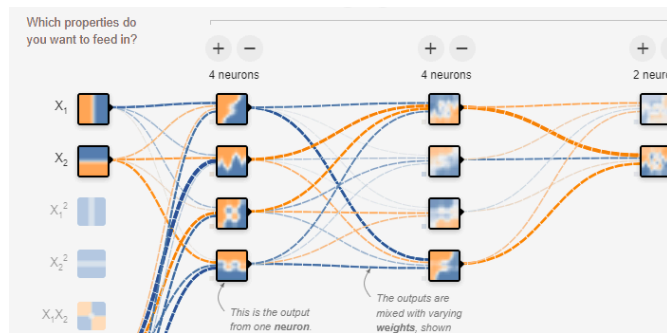


Basic Terms



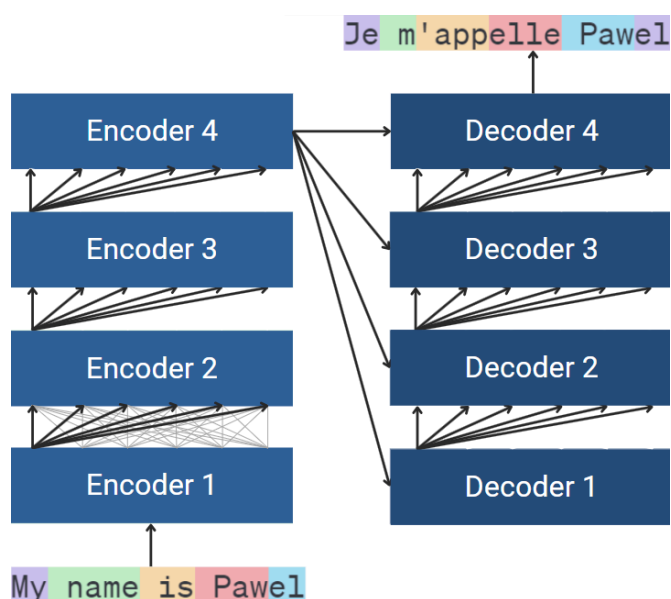
- **Supervised Learning:** Model is trained on a labeled dataset (input + correct output). This is the basis for fine-tuning.
- **Unsupervised Learning:** Model learns from data without labels, discovering patterns and structures independently.
- **Reinforcement Learning:** Model interacts with an environment, receiving rewards or penalties to learn (e.g., AlphaGo).
- **Deep Learning:** Using neural networks with many layers. Can be used in supervised, unsupervised, and reinforcement learning.
- **Neural Networks:** Computational models inspired by the human brain, consisting of layers of interconnected nodes ("neurons") that process data.
- **Transformers:** Architecture for sequential data, using self-attention to weigh the importance of different input parts.
- **Large Language Models (LLMs):** Trained on large datasets to generate and understand human-like text. Combine supervised, unsupervised, and deep learning.

Neural Networks



- **Weights of a model:** Neurons' connection strength
- **Activation function:** Determines how each neuron responds to its inputs.
- **Loss function:** evaluates the model's accuracy by comparing the predicted outputs to the actual data.
- **Backpropagation:** Updating weights to reduce overall loss.
- **Recommended tools:** [TensorFlow Playground](#).

Transformers



- **Strengths:** Transformers revolutionized AI by effectively handling sequential data like text and speech.
- **Tokenization:** The input data (like text) is split into tokens.
- **Positional encoding:** Encoding a position to each token so the model can understand, e.g., the order of words.
- **Self-attention:** Used by encoders and decoders. It allows each node to interact with all other nodes.
- **Feedforward network:** Passes the data to the next layer.
- **Masked attention:** Prevents future tokens from being seen during generation.
- **Recommended tools:** [LLM Visualization](#), [OpenAI Tokenizer](#)